

HMDA 2024

RESPONSIBLE ML CAPSTONE

XGBoost Mortgage Approval Model

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MAIN REPORT QUESTION

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x x x
x x x
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x x x



What is the **optimization objective**?



What are the known **failure modes**?



How are **subgroup errors** measured?



What **risks** remain after mitigation?



Is **deployment defensible**?



What is the optimization objective?

Binary classification task: Approve or Deny a mortgage application

Filtered data
4,354,586
rows

Component	Our Choice	Rationale
Main optimization metric	AUC	The dataset: Approvals > Denials. AUC better captures ranking quality.
Training objective	Log-loss	The quality of predicted approval probabilities.
Fairness threshold	AIR < 0.80, FPR, FNR	Flags potential disparate impact.

Stakeholders

Applicants	Lenders / Loan Review Teams	Protected / Minority Groups
Avoid false denial and unfair credit loss	Reliable approval probability ranking	Detect lower selection rates or higher error rates

Features

- **Numeric_features**
property_value
income
tract_population
ffiec_msa_md_median_family_income
tract_to_msa_income_percentage
- **Categorical_features**
state_code
derived_loan_product_typed
erived_dwelling_category
loan_purpose
lien_status
occupancy_type





What is the optimization objective?

- XGBoost achieved the strongest overall ranking and probability performance.
- Random Forest had lower FNR, but weaker AUC/log-loss and higher FPR.
- XGBoost serves as the main model in our audit.

Model	Accuracy	AUC	Log Loss	Precision	Recall	FPR	FNR
Logistic Regression	0.74	0.74	0.51	0.76	0.93	0.77	0.05
XGBoost	0.78	0.80	0.46	0.80	0.93	0.60	0.06
Random Forest	0.76	0.78	0.49	0.77	0.96	0.76	0.03

XGBoost Confusion Matrix		
Actual Label	Denied (0)	Approved (1)
	Denied (0)	Approved (1)
Denied (0)	TN 103,647	FP 165,839
Approved (1)	FN 47,345	TP 660,470





What are the known failure modes?

Failure Type	Trigger	Affected Group	Severity
Proxy Discrimination	Tract income features in SHAP top 10 (flagged as proxies)	Minority applicants	High
Adversarial Gaming	Loan purpose and income are top SHAP drivers; both are known in HMDA applications	All applicants	High
Feature Drift	AUC drops from 0.68 to 0.56 across prediction buckets	Borderline applicants	High
Fairness Disparity	Black AIR = 0.92; higher FNR vs White Applicants	Black Applicants	High



What are the known failure modes?

Manipulation Risk

- Income and loan purpose drive predictions
- Susceptible to probing and strategic input changes

Fairness Risk

- Tract-level features act as demographic proxies
- Creates disparity without using race directly

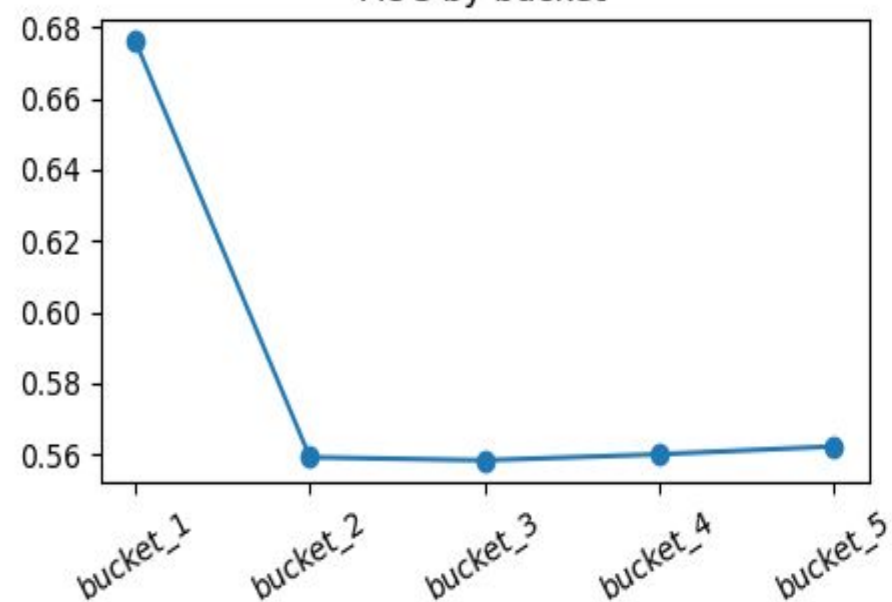
Boundary Weakness

- Model performs worst near decision threshold
- Errors concentrated on borderline applicants

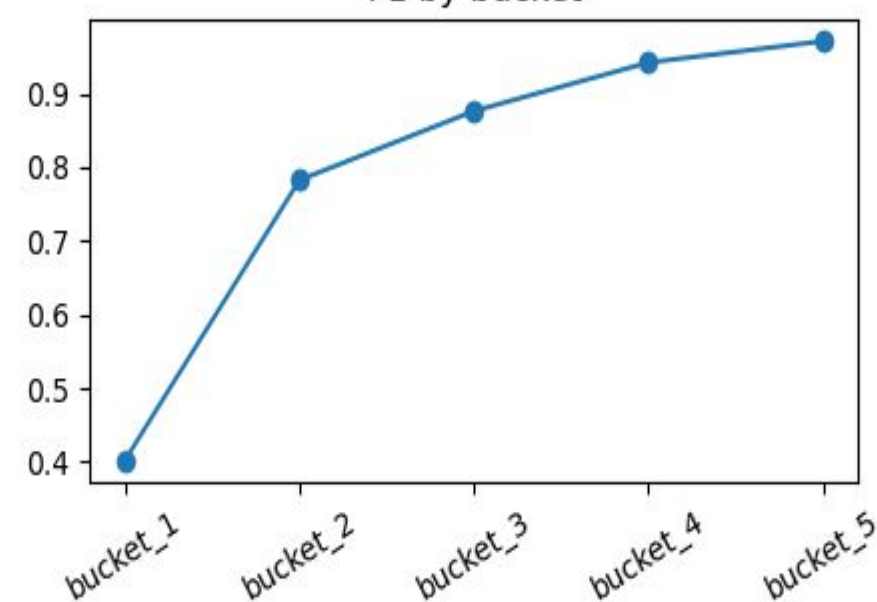
Limited Actionability

- Counterfactuals require unrealistic changes
- Denied applicants cannot act on model feedback

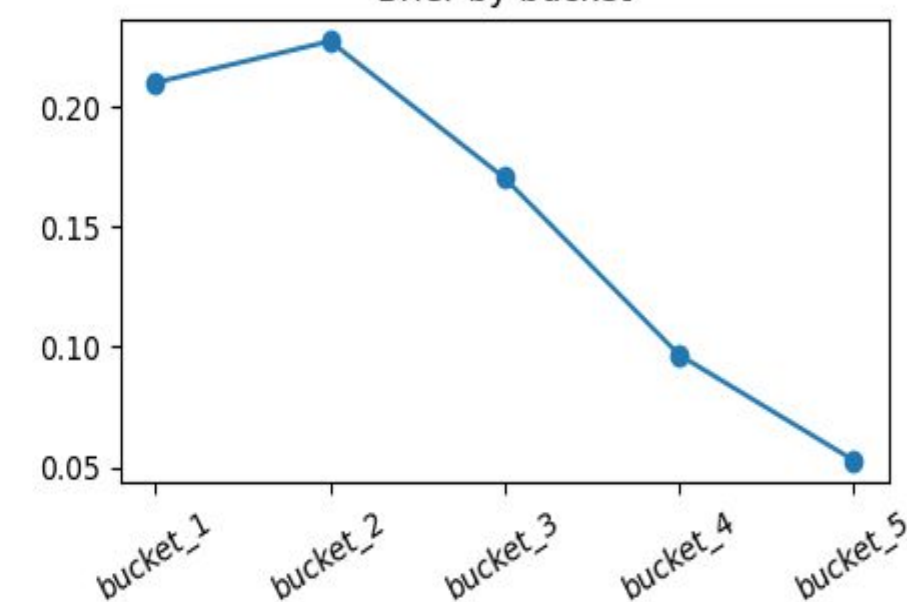
AUC by bucket



F1 by bucket



Brier by bucket





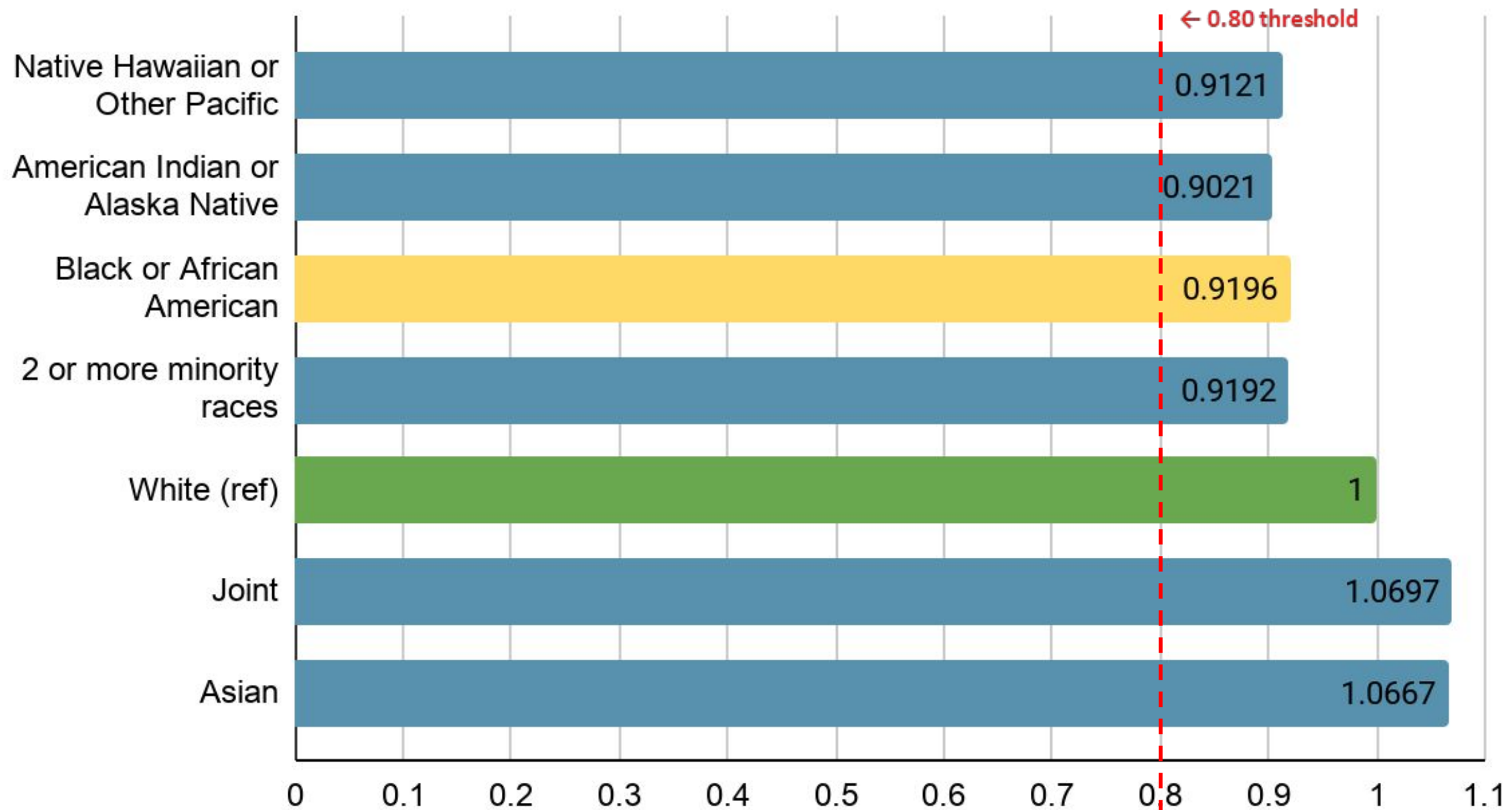
How are subgroup errors measured?

AIR Adverse Impact Ratio	ME Marginal Effect	SMD Standardized Mean Difference
Approval rate of a protected group ÷ Approval rate of the reference group	Absolute difference in selection rates between a group and the reference group	Gap in predicted probability scores between groups, measured in standard deviations .
Reference groups: White, Male, Not Hispanic or Latino	Captures absolute gap in percentage points	Gap in predicted probability scores, measured in standard deviations
Four-Fifths Rule: $AIR < 0.80 \Rightarrow$ potential fair-lending violation	Negative = disadvantaged group	Cohen's d: small <0.2 , medium <0.5 , large <0.8



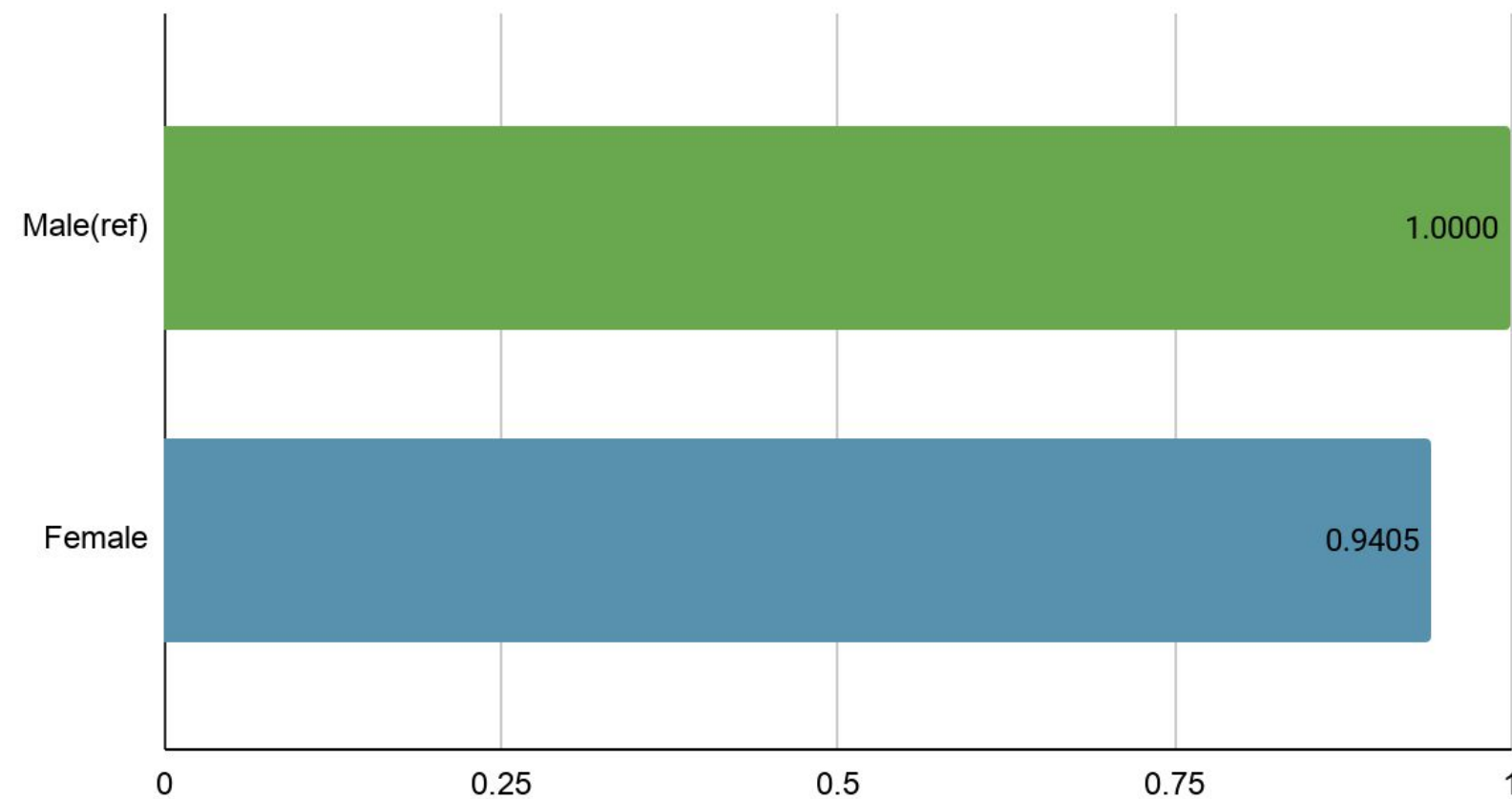
How are subgroup errors measured?

AIR by Race (reference group: White)

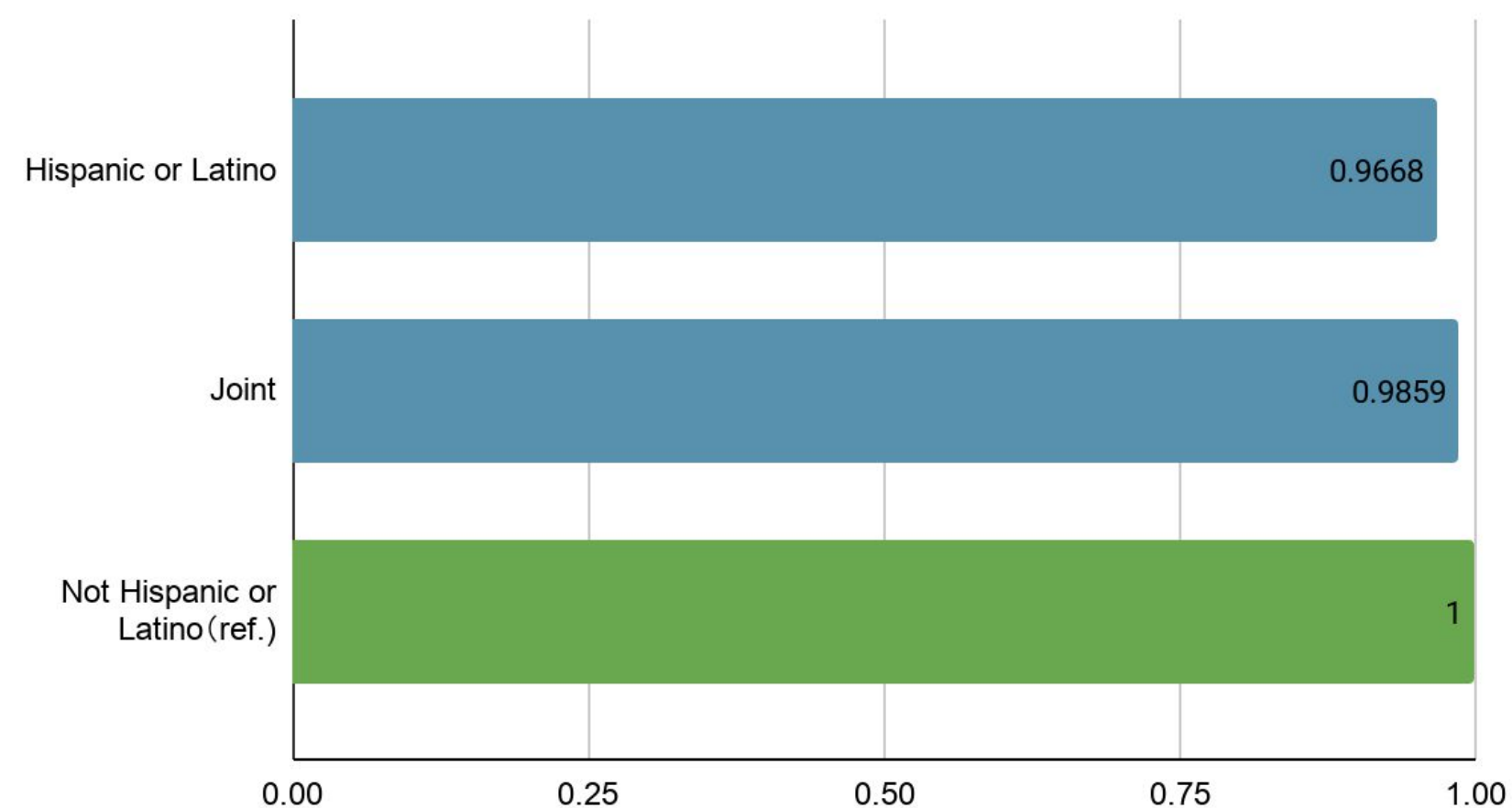


How are subgroup errors measured?

AIR by Sex (reference group: Male)



AIR by Ethnicity (reference group: Not Hispanic or Latino)

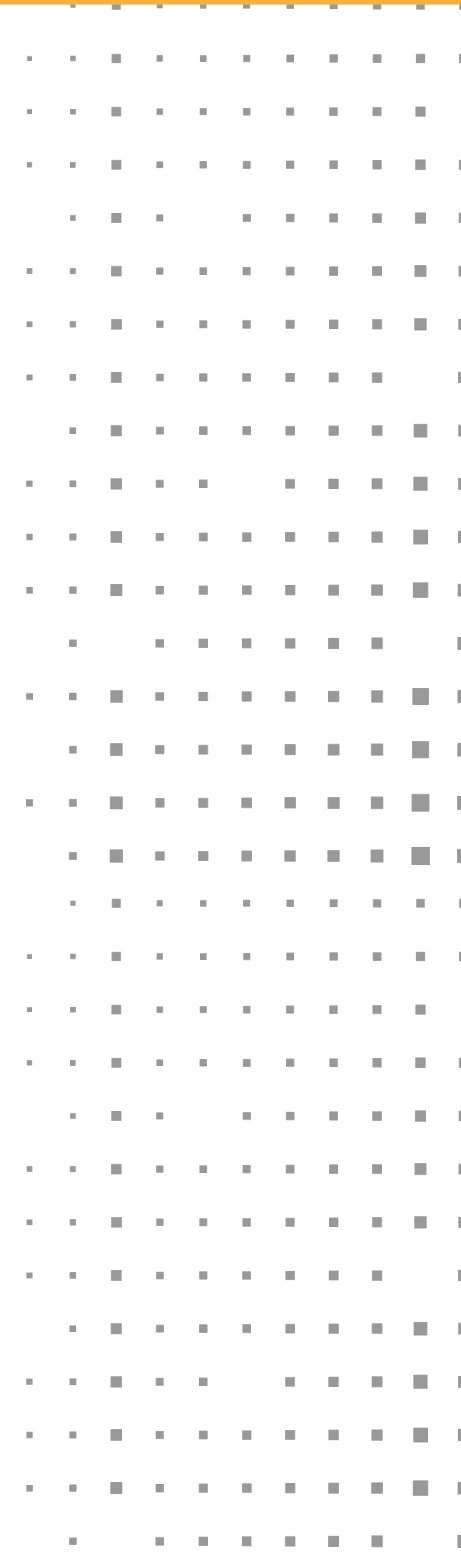




How are subgroup errors measured?

AIR by Race x Sex Subgroup (reference group: White/Male)

subgroup	n	AIR
American Indian or Alaska Native / Female	4141	0.835103
Native Hawaiian or Other Pacific Islander / Female	1127	0.848078
2 or more minority races /Female	1594	0.861498
Black or African American / Female	63840	0.873907
Native Hawaiian or Other Pacific Islander / Male	2086	0.916927
American Indian or Alaska Native / Male	6311	0.914287
2 or more minority races / Male	1906	0.931712
Black or African American / Male	57572	0.932012
White / Female	255159	0.945140
White/Male	404201	1.000000
Joint /Female	1162	1.020806
Asian / Female	23948	1.029299
Asian / Male	46907	1.051585
Joint/Male	1350	1.069586





The remaining risks

Hidden Subgroup Problems	Sensitivity to Data Change	Small manipulations changes Robustness
- A group may look “fine” when you check race alone... but becomes unfair when you check race * sex together. - This means the model has pockets of harm that aren’t obvious at first glance.	- Small shifts in income, credit score, or applicant mix can change fairness outcomes. - Fairness breaks faster than accuracy – the model can look “good” overall while harming certain groups.	- Tiny changes in inputs can push borderline applicants into denial. - - This risk is higher for minority groups.



Is deployment defensible?

Q5 — Deployment Decision

Deployment Decision

CONDITIONAL DEPLOY

Why deploy:

XGBoost achieves AUC 0.7977, accuracy 0.7819, and best log-loss (0.4654) across all three models. The generalization gap is only -0.0026 , confirming minimal overfitting. Stress tests show minimal performance sensitivity (max AUC drop of 0.005), with all PSI values stable.

Top SHAP drivers — loan purpose, income, and property value — are financially grounded and defensible under HMDA.

Conditions:

Deploy as a decision-support tool only, not for sole automated decision-making. Fairness gaps and security controls must be addressed before full deployment.

Monitoring Plan

Performance (monthly)

Track AUC, FPR, and FNR. Trigger retraining if AUC drops below 0.75 or FPR rises above 0.70.

Fairness (quarterly)

Monitor AIR by race, sex, and ethnicity using a policy threshold of 0.80. Flag any group falling below this for immediate review. Priority groups: Black or African American (AIR 0.851) and Native Hawaiian/Pacific Islander (AIR 0.834).

Drift (monthly)

Monitor PSI on all input features. Trigger retraining if any feature PSI exceeds 0.25. Current PSI values are all stable.

Security (weekly)

Log all API queries and flag unusual submission patterns as signals of gaming or model extraction attempts, especially given the concentration of importance in a few features (income, loan purpose, property value).

Risk Acceptance

Accepted

Strong generalization (AUC gap -0.0026), stable PSI across all features, and minimal performance degradation under stress testing (max AUC drop 0.005) are sufficient for conditional, monitored deployment.

Accepted with Conditions

Fairness gaps — Black or African American AIR 0.851, Native Hawaiian/Pacific Islander AIR 0.834, and intersectional subgroups below 0.80 — along with proxy feature risks (e.g., geographic and income-related variables) are accepted temporarily, subject to quarterly review and regulatory disclosure.

Not Accepted

Sole automated decision-making without human review. Rate limiting, query logging, input validation, and pipeline integrity checks must be in place before go-live.

Audit Record The project notebook, hosted on GitHub, serves as the formal governance artifact, documenting all model decisions, thresholds, fairness findings, and risk assessments end to end.



THANK YOU

